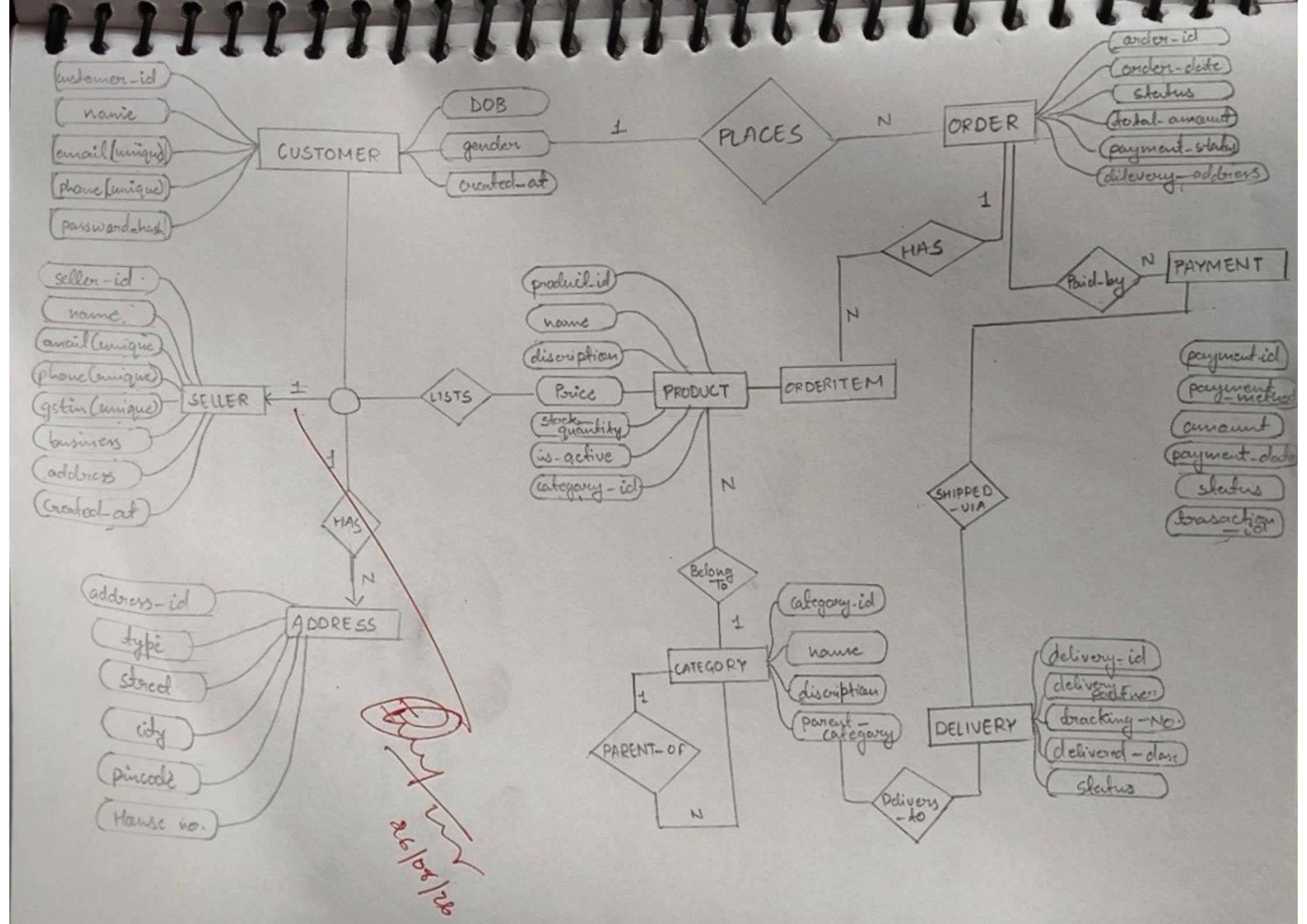

customer_id
name
email (unique)
phone (unique)
password_hash
CUSTOMER
DOB
gender
created_at
1
PLACES
N
ORDER
order_id
order_date
status
total_amount
payment_status
delivery_address
1
HAS
N
Paid_by
N
PAYMENT
payment_id
payment_method
amount
payment_date
status
transaction_id
seller_id
name
email (unique)
phone (unique)
gstin (unique)
business
address
created_at
SELLER
1
LISTS
product_id
name
discription
Price
stock_quantity
is_active
category_id
PRODUCT
ORDERITEM
1
HAS
N
address_id
type
street
city
pincode
House no.
ADDRESS
N
Belong To
1
category_id
name
discription
parent_category
CATEGORY
1
PARENT_OF
N
SHIPPED_VIA
DELIVERY
delivery_id
delivery_partner
tracking_no.
delivered_date
status
Delivery_to
26/04/26

# EXPERIMENT-2

Aim:

To convert the ER diagram of an Indian E-commerce platform into a relational schema & implement it using MySQL with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. To insert sample data and demonstrate referential integrity violations.

Source Code:

```
CREATE DATABASE Ecommerce DB;
USE Ecommerce DB;
CREATE TABLE Customer C
    Customer-ID INT PRIMARY KEY,
    first-NAME VARCHAR(30) NOT NULL,
    Middle-name VARCHAR(30), NOT NULL,
    Last-name VARCHAR(30) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15) NOT NULL UNIQUE,
    House VARCHAR(50),
    Pin VARCHAR(10),
    City VARCHAR(50)
);
```

```
CREATE TABLE Seller (
    Seller_ID INT PRIMARY KEY,
    Seller_Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15) NOT NULL UNIQUE
    );
CREATE TABLE Category (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50) NOT NULL UNIQUE,
    Description VARCHAR(200)
    );
CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    Stock INT NOT NULL,
    Category_ID INT,
    Seller_ID INT,
FOREIGN KEY (Category_ID)
    REFRENCE Category (Category_ID)
    ON DELETE SET NULL,
```

```
    FOREIGN KEY (Seller_ID)
        REFERENCES Seller (Seller_ID)
        ON DELETE SET NULL
    );

CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    Date DATE NOT NULL,
    Amount DECIMAL (10,2) NOT NULL,

  FOREIGN KEY(Customer_ID)
    REFERENCES Customer (Customer_ID)
    ON DELETE CASCADE
    );

CREATE TABLE OrderItem (
    Order_ID INT,
    Product_ID INT,
   PRIMARY KEY (Order_ID, Product_ID),
  FOREIGN KEY (Order_ID)
    REFRENCES Orders (Order_ID)
    ON DELETE CASCADE,
 FOREIGN KEY (Product_ID)
    REFRENCE Product (Product_ID)
    ON DELETE CASCADE
```

```
);

CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Amount DECIMAL(10,2) NOT NULL,
    Method VARCHAR(30) NOT NULL,
    Status VARCHAR(30) NOT NULL,

    FOREIGN KEY (Order_ID)
        REFRENCES Orders(Order_ID)
        ON DELETE CASCADE
    );

CREATE TABLE Delivery (
    Delivery_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Delivery_Date DATE,
    Status VARCHAR(30) NOT NULL,
    Tracking VARCHAR(50) UNIQUE,

FOREIGN KEY(Order_ID)
    REFRENCES Orders (Order_ID)
    ON DELETE CASCADE
);
```

## INSERT SAMPLE DATA

### Customer

```
INSERT INTO Customer
VALUES
(1, 'Anushka', 'anushka@gmail.com', '7753812665', '12A',
'123019', 'Lucknow');

(2, 'Yashi', 'yashi@gmail.com', '9958127027', '45B'
,'760001', 'Delhi');
```

### Seller

```
INSERT INTO Seller
VALUES
(101, 'Tech world', 'tech@gmail.com', '9811100000'),
(102, 'Fashion Hub', 'fashi@gmail.com', '98222222222
```

### Category

```
INSERT INTO CATEGORY
VALUES
(201, 'Electronics', 'Electronic Products');

(202, 'Fashion', 'Fashion Products');
```

Product

```
INSERT INTO Product
 VALUES
(301, 'LAPTOP', 65000.00, 10, 201, 101),
(302, 'SM artphone', 30000.00, 20, 201, 101),
(303, 'Jacket', '2500.00', 15, 202, 102);
```

Orders

```
INSERT INTO Orders
VALUES
(401, 1, '2026-08-19', 65000.00),
(402, 2, '2026-08-19', 30000.00);
```

Order Item

```
INSERT INTO Order Item
 VALUES
(401, 301),
(402, 302);
```

Payment

```
INSERT INTO Payment
 VALUES
```

(501, 401, 65000.00, 'UPI', 'Successful'),
(502, 402, 30000.00, 'Card', 'Successful');

Delivery

INSERT INTO Delivery
VALUES
(601, 401, '2026-08-22', 'In Transit', 'TRK10001')
(602, 402, '2026-08-23', 'Processing', 'TRK10002');

Demonstration of Refrential integrity violation

1) Foreign key violation

Trying to insert an order for a customer who does not exist:

INSERT INTO ~~ORDERS~~ Orders
VALUES
(403, 999, '2026-08-19', 5000.00)

Output

ERROR 1452 (23000):

Cannot add or update a child row:
a foreign key constraint fails.

Here, Customer-ID = 999 doesn't exist in the customer table.

2. Primary Key violation

```
INSERT INTO Customer
VALUES
(1, 'Rahul', 'rahul@gmail.com', '9876543210',
'20A', '110002', 'Delhi');
```

Output

```
ERROR 1062 (23000):
Duplicate entry (1) for key 'PRIMARY'.
```

Because Customer-ID = 1 already exists.

UNIQUE Constraint Violation

```
INSERT INTO Customer
VALUES
(3, 'Rahul', 'rahul@gmail.com', '9876543211', '20A'
'110002', 'Delhi');
```

Output

```
ERROR 1062 (23000):
Duplicate entry 'pari@gmail.com' for key 'email'
```

Because the email address is declared as UNIQUE.

# Demonstration of ON DELETE CASCADE

First, delete customer1:

```
DELETE FROM Customer
WHERE Customer_ID = 1;
```

Because orders Customer_ID has ON DELETE CASCADE, all ~~ordering~~ orders belonging to customer 1 will also be deleted.

The related OrderItem, Payment & Delivery records will also be deleted through their cascading foreign keys.

# Demonstration of ON DELETE SET NULL

Delete a Seller:

```
DELETE FROM Seller
WHERE Seller_ID=101:
```

Since Product.Seller_ID uses ON DELETE SET NULL the seller information will be removed but the products will remain.

Check the Product table:

```
SELECT Product_ID, Name, Seller_ID
FROM Product.
```

Output

| Product_ID | Name | Seller_ID |
|---|---|---|
| 301 | Laptop | NULL |
| 302 | Smartphone | NULL |
| 303 | Jacket | 102 |

RELATIONAL SCHEMA

Customer (
Customer_ID PK,
Name,
Email, UNIQUE,
Phone UNIQUE,
House,
Pin,
City
)

Seller (
Seller_ID PK,

Seller - Name,
Email UNIQUE
Phone UNIQUE
)

Category (
Category_ID PK,
Category_Name UNIQUE,
)
Description

Product (
Product_ID PK,
Name,
Price,
Stock,
Category_ID FK,
Seller_ID FK,
)

Orders (
Order_ID PK,
Customer_ID FK,
Date,
Amount.
)

Order Item (

Order-ID PK, FK,

Product-ID PK, FK

)

Payment (

Payment-ID PK,

Order-ID FK UNIQUE,

Amount,

Method,

Status

)

Delivery (

Delivery-ID PK,

Order-ID FK UNIQUE,

Delivery-Date,

Status

Tracking UNIQUE

)

OUTPUT

The given ER diagram is successfully converted into a relational schema. All tables are created successfully with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE & ON DELETE SET NULL constraints. Sample data is inserted successfully & referential integrity violations are demonstrated.

26/08/26